FUN-6.D.1 Substitution of variables is a technique for finding antiderivatives.
FUN-6.D.2 For a definite integral, substitution of variables requires corresponding changes to the limits of integration.

6-9 Integration With U-Substitution

The 5 INTEGRATION RULES YOU MUST KNOW STONE COLD	
$\int \sin u \ du = -\cos u + C$	$\int \cos u \ du = \sin u + C$
$\int e^u \ du = e^u + C$	$\int u^{-1} \ du = \int \frac{1}{u} \ du = \ln\lvert u\rvert + C$
$\int u^n \ du = \frac{1}{n+1} u^{n+1}$	(where $n \neq -1$)

Note: Each of the integration rules above uses the variable u along with du instead of x and dx. This is because the problems we encounter on the AP Exam rarely involve ***just*** x ... instead, we will see expressions involving x.

Because of this, we have to be careful not to misapply our integration rules!

$$\int \cos(x)dx = \sin(x) + C\,, \quad \text{but} \int \cos(2x)\,dx \neq \sin(2x) + C$$

Why?

$$\frac{d}{dx}[\sin(2x)] = 2\cos(2x) \text{ by applying the chain rule.}$$

This concept is essentially the same idea as using **<u>the chain rule</u>** for derivatives.

Given $y = F(u)$ and $u = g(x)$, then

$$\frac{d}{dx}[F(u)] = \frac{d}{dx}\left[F\big(g(x)\big)\right] = F'(g(x)) \cdot g'(x)$$

So, from the definition of an antiderivative, it follows that:

$$\int F'\big(g(x)\big) \cdot g'(x)\, dx = F\big(g(x)\big) + C = F(u) + C$$

To find antiderivatives of problems like this, we will use a method called **<u>u-substitution</u>**.

The first and most important step in using u substitution is identifying the "u" in the expression.

Identifying "u"	• "u" is usually the "inside function". • It is usually the stuff in the **parentheses**. • It is a more **complicated** expression that takes the place of where x would normally be.

Strategies for u Substitution
1. Identify u. Set an expression equal to u (look for parenthesis or radicals). 2. Find du (differentiate both sides of your u equation). 3. Substitute u into the original problem. 4. Fix any constants needed to make the du work. 5. Integrate using an integral formula. 6. Substitute back to an expression for x (only for indefinite integrals).

Evaluate the following integrals:

Example 1: $\int 2x(x^2+3)^7\,dx =$

$u = x^2+3$

$\frac{du}{dx} = 2x$

$du = 2x\,dx$

$= \int u^7\,du$

$= \frac{1}{8}u^8 + C$

$= \frac{1}{8}(x^2+3)^8 + C$

Example 2: $\int 3x(7x^2-2)^4\,dx =$

$u = 7x^2 - 2$

$du = 14x\,dx$

$= \frac{3}{14}\int 14x(7x^2-2)^4\,dx$

$= \frac{3}{14}\int u^4\,du$

$= \frac{3}{14}\cdot\frac{1}{5}u^5 + C$

$= \boxed{\frac{3}{70}(7x^2-2)^5 + C}$

Example 3: $\int 2\sin(5x)\ dx =$

$u = 5x$

$du = 5dx$

$= \frac{2}{5}\int 5\sin(5x)\ dx$

$= \frac{2}{5}\int \sin u\, du$

$= -\frac{2}{5}\cos u + C = -\frac{2}{5}\cos(5x) + C$

Example 4: $\int \frac{x^2}{\sqrt{x^3+1}}\,dx =$

$u = x^3 + 1$

$du = 3x^2\,dx$

$= \frac{1}{3}\int 3x^2 \left(x^3+1\right)^{-1/2} dx$

$= \frac{1}{3}\int u^{-1/2}\,du$

$= \frac{1}{3} \cdot 2u^{1/2} + C$

$= \frac{2}{3}\left(x^3+1\right)^{1/2} + C$

MC: $\frac{2\sqrt{x^3+1}}{3} + C$

Example 5: $\int xe^{x^2-1}\,dx =$

$u = x^2 - 1$

$du = 2x\,dx$

$= \frac{1}{2}\int 2x\, e^{(x^2-1)}\, dx$

$= \frac{1}{2}\int e^u\, du$

$= \frac{1}{2}e^u + C = \frac{1}{2}e^{(x^2-1)} + C$

$\frac{e^{x^2-1}}{2} + C$

Example 6: $\int \frac{3}{4-2x}\,dx =$

$u = 4-2x$

$du = -2dx$

$$\int 3 \cdot \frac{1}{4-2x}\,dx$$

$$= \frac{3}{-2}\int \frac{1}{4-2x} \cdot -2\,dx$$

$$= \frac{-3}{2}\int \frac{1}{u}\,du$$

$$= \frac{-3}{2}\ln|u| + C = \frac{-3}{2}\ln|4-2x| + C$$

Example 7: $\int \frac{-4}{(5+3x)^2}\,dx =$

$u = 5+3x$

$du = 3\,dx$

$$= \frac{-4}{3}\int \frac{1 \cdot 3}{(5+3x)^2}\,dx$$

$$= -\frac{4}{3}\int \frac{1}{u^2}\,du$$

$$= -\frac{4}{3}\int u^{-2}\,du$$

$$= -\frac{4}{3}\cdot -1\,u^{-1} + C = \frac{4}{3}(5+3x)^{-1} + C$$

$$= \frac{4}{3(5+3x)} + C$$

Example 8: $\int [\sin(3x) - \cos(x)]dx =$

$u = 3x$
$du = 3dx$

$= \frac{1}{3}\int 3\sin(3x)\,dx - \int \cos x\,dx$

$= \frac{1}{3}\int \sin u\,du - \sin x + C$

$= -\frac{1}{3}\cos u - \sin x + C$

$= -\frac{1}{3}\cos(3x) - \sin x + C$

Example 9: $\int \tan x \; dx =$

$u = \cos x$

$du = -\sin x \, dx$

$$= \int \frac{\sin x}{\cos x} \, dx$$

$$= -\int -\sin x \cdot \frac{1}{\cos x} \, dx$$

$$= -\int \frac{1}{u} \, du$$

$$= -\ln|u| + C$$

$$= \boxed{-\ln|\cos x| + C}$$

$a \cdot \ln b = \ln b^a$

$$= \ln\left|(\cos x)^{-1}\right| + C$$

$$= \ln\left|\frac{1}{\cos x}\right| + C$$

$$= \boxed{\ln|\sec x| + C}$$

Every definite integral has 3 important components:

$$\int_a^b f(x)\,dx$$

1. The integrand $f(x)$.
2. The differential dx.
3. The limits of integration a and b

All three components must be in terms of the **SAME** variable. So, if we use u substitution with a definite integral, we must also remember to change the **limits of integration**.

AP EXAM TIP: With definite integrals, we do **NOT** want to convert back to x after integrating. It is much easier to evaluate the antiderivative in terms of u!

Evaluate the integrals:

Example 10: $\int_0^1 2x(x^2+4)^3\,dx =$

$u = x^2 + 4$ *

$du = 2x\,dx$

$u(1) = 1^2 + 4 = 5$

$u(0) = 0^2 + 4 = 4$

$$= \int_4^5 u^3\,du$$

$$= \frac{1}{4}u^4 \Big|_4^5$$

$$= \frac{1}{4}(625 - 256) = \frac{1}{4}(369) = \frac{369}{4}$$

OR $\frac{625}{4} - \frac{256}{4} = \frac{369}{4}$

Example 11: $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sin(2x)\ dx =$

$u = 2x$

$du = 2\,dx$

$u\left(\frac{\pi}{2}\right) = 2 \cdot \frac{\pi}{2} = \pi$

$u\left(\frac{\pi}{6}\right) = 2 \cdot \frac{\pi}{6} = \frac{\pi}{3}$

$$\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 2\sin(2x)\,dx$$

$$= \frac{1}{2}\int_{\frac{\pi}{3}}^{\pi} \sin u\,du$$

$$= -\frac{1}{2}\cos u \Big|_{\frac{\pi}{3}}^{\pi} = -\frac{1}{2}\left(\cos\pi - \cos\frac{\pi}{3}\right)$$

$$-\frac{1}{2}\cos\pi - -\frac{1}{2}\cos\frac{\pi}{3}$$

$$= -\frac{1}{2}\left(-1 - \frac{1}{2}\right)$$

$$= -\frac{1}{2}\left(\frac{-3}{2}\right) = \boxed{\frac{3}{4}}$$

Example 12: $\int_1^e \frac{\ln x}{x}\,dx =$

$$\int_1^e \ln x \cdot \frac{1}{x}\,dx$$

$$= \int_0^1 u\,du$$

$$= \frac{1}{2}u^2\Big|_0^1$$

$$= \frac{1}{2} - 0 = \frac{1}{2}$$

$u = \ln x$

$du = \frac{1}{x}\,dx$

$u(e) = \ln e = 1$

$u(1) = \ln 1 = 0$

You do not have to change the limits of integration. You can choose to either "back substitute" your expression in terms of x and keep the original boundaries OR you can keep your answer in terms of u and change the boundaries. The choice is up to you. **The AP Exam likes to ask questions which require you to change the boundaries.**

Example 13:

By using the substitution $u = 2\sin x$, which of the following is equivalent to $\int_0^{\frac{\pi}{2}} \cos x \sqrt{2\sin x}\ dx$?

A. $\cancel{\int_0^{\frac{\pi}{2}} u^{\frac{1}{2}}\, du}$ B. $\frac{1}{2}\int_0^{\frac{\pi}{2}} u^{\frac{1}{2}}\, du$ C. $\cancel{\int_0^{2} u^{\frac{1}{2}}\, du}$ D. $\frac{1}{2}\int_0^{2} u^{\frac{1}{2}}\, du$

$\frac{1}{2}\int_0^{\pi/2} 2\cos x\sqrt{2\sin x}\ dx$

$= \frac{1}{2}\int \sqrt{u}\, du$

$u = 2\sin x$

$du = 2\cos x\, dx$

$u\left(\frac{\pi}{2}\right) = 2\sin\frac{\pi}{2} = 2$

Change of Variable Method

When we do the "u-substitution" method, we can never insert variables into the integral. We have to work with whatever we are given. What happens when we can't just insert a constant to make du? Sometimes, we won't be able to integrate, but sometimes we can use a "change of variable" method. This is still a "substitution", but we will go about it a little differently than our normal "u-substitution" process.

Example 14: Evaluate: $\int x\sqrt{2x-1}\,dx$

$u = 2x - 1$

$du = 2\,dx$

$u = 2x - 1$

$u + 1 = 2x$

$\frac{u+1}{2} = x$

$\frac{1}{2}u + \frac{1}{2} = x$

$$\frac{1}{2}\int 2x\sqrt{2x-1}\,dx$$

$$= \frac{1}{2}\int \left(\frac{1}{2}u + \frac{1}{2}\right)\sqrt{u}\,du$$

$$= \frac{1}{2}\int \left(\frac{1}{2}u + \frac{1}{2}\right)u^{1/2}\,du$$

$$= \frac{1}{2}\int \left(\frac{1}{2}u^{3/2} + \frac{1}{2}u^{1/2}\right)du$$

$$= \frac{1}{2}\left[\frac{1}{2}\cdot\frac{2}{5}u^{5/2} + \frac{1}{2}\cdot\frac{2}{3}u^{3/2}\right] + C$$

$$= \frac{1}{10}(2x-1)^{5/2} + \frac{1}{6}(2x-1)^{3/2} + C$$

Integration involving Inverse Trigonometric Functions

This sounds as if we are going to learn how to take integrals of expressions that contain the inverse trigonometric functions. When, in fact, we are going to integrate expressions that result in answers that contain inverse trigonometric functions. The good news is that there are only three forms with which we need to concern ourselves.

Integrals Involving Inverse Trigonometric Functions	
Let u be a differentiable function of x and $a > 0$	
Basic Form	**U-Substitution Form**
$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$	$\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\left(\frac{u}{a}\right) + C$
$\int \frac{dx}{1+x^2} = \arctan x + C$	$\int \frac{du}{a^2+u^2} = \frac{1}{a}\arctan\left(\frac{u}{a}\right) + C$
$\int \frac{dx}{x\sqrt{x^2-1}} = \operatorname{arcsec}\lvert x\rvert + C$	$\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\operatorname{arcsec}\left\lvert\frac{u}{a}\right\rvert + C$

Notice you don't have to memorize both sets of formulas. You can use the U-substitution formulas with $a = 1$ to get the basic formulas.

Example 15: $\int \frac{dx}{\sqrt{4-x^2}}$

$$\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\left(\frac{u}{a}\right)+C$$

$u^2 = x^2$ $\quad a^2 = 4$

$u = x$ $\quad a = 2$

$du = 1dx$

$= \arcsin\left(\frac{x}{2}\right)+C$

Example 16: $\int \frac{dx}{\boxed{2}+\boxed{9x^2}}$

$$\int \frac{du}{a^2+u^2} = \frac{1}{a}\arctan\left(\frac{u}{a}\right)+C$$

$u^2 = 9x^2$

$u = 3x$

$du = 3dx$

$a^2 = 2$

$a = \sqrt{2}$

$$\frac{1}{3}\int \frac{3dx}{2+9x^2} = \frac{1}{3}\cdot\frac{1}{\sqrt{2}}\arctan\left(\frac{3x}{\sqrt{2}}\right)+C$$

$$= \frac{1}{3\sqrt{2}}\arctan\left(\frac{3x}{\sqrt{2}}\right)+C$$

Example 17: $\int \frac{dx}{x\sqrt{\boxed{4x^2}-\boxed{9}}}$

$$\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a}\operatorname{arcsec}\left|\frac{u}{a}\right| + C$$

$u^2 = 4x^2$

$u = 2x$

$du = 2dx$

$a^2 = 9$

$a = 3$

$$2 \cdot \frac{1}{2}\int \frac{2dx}{2x\sqrt{4x^2-9}} = \frac{1}{3}\operatorname{arcsec}\left|\frac{2x}{3}\right| + C$$